**Module 6: Flattening and Fully Connected Layers**

**1. What happens after convolution?**

**After multiple convolutional and downsampling layers, the CNN has learned feature maps representing patterns in the image.**

**For a traditional classification architecture, these features are passed to a classification head.**

**2. Flattening**

**Flattening converts a multidimensional feature tensor into a one-dimensional vector.**

**For example:**

**Original feature map:**

**1 2**

**3 4**

**After flattening:**

**[1, 2, 3, 4]**

**Flattening changes the arrangement of the data, not its numerical values.**

**It does not itself learn features.**

**3. Fully connected layer**

**A fully connected layer is also called a dense layer.**

**Each output neuron receives input from every value in the preceding layer.**

**A simplified equation is:**

**y = f(Wx + b)**

**Where:**

- **x = input vector**
- **W = trainable weight matrix**
- **b = bias**
- **f = activation function**

**The layer learns combinations of extracted features that help produce predictions.**

**4. Output layer**

**Suppose a model classifies images into two categories:**

- **Cat**
- **Dog**

**The output layer can produce two class scores.**

**A softmax function may convert these scores into values that sum to 1.**

**For binary classification, another common design uses one output logit followed by a sigmoid function.**

**The correct output structure depends on the task.**

**5. Why do we not directly use ANN on the original image?**

**A basic fully connected ANN can process flattened images, but it may require many parameters and does not naturally exploit the local spatial structure of images.**

**CNN first extracts useful visual features and then uses a classification head to make the prediction.**

**6. Is flattening always necessary?**

**No.**

**Some modern CNNs use global average pooling instead.**

**Global average pooling calculates the average value of each feature map, producing one value per channel.**

**Feature Maps**

**|**

**v**

**Global Average Pooling**

**|**

**v**

**One Value Per Channel**

**|**

**v**

**Output Layer**

**This can reduce the number of parameters in the classifier.**

**Conclusion**

**Flattening reshapes the extracted features into a vector. Fully connected layers use these features to predict the output. Some modern CNNs use global average pooling instead of flattening.**